Field Station Rajasthan · 30 October 2026

λ Logos GRAMA

# Pramana

*pramāṇa (Sanskrit): a valid means of knowing*

Any claim, traced to the numbers it rests on.

Amruth Deepak Bhat

Built on Logos

# How much water does one AI answer use?

# Three published answers

## 519 mL

**A bottle, per 100-word email**

Ren et al., Making AI Less "Thirsty"

> sumption. Additionally, GPT-3 needs to "drink" (i.e., consume) a **500ml bottle of water** for roughly 10 – 50 medium-length responses, depending on when and where it is deployed.

Washington Post, Sept 2024, using the method of this UC Riverside paper

## 0.26 mL

**Five drops, per Gemini prompt**

Measuring the environmental impact of delivering AI at Google scale

> Gemini Apps text prompt over one year. We identify that the median Gemini Apps text prompt use
> television (0.24 Wh) and consumes the equivalent of five drops of water (0.26 mL). While these in
> activities, reducing the environmental impact of AI serving continues to warrant important attentior

Elsworth et al., Google, arXiv, Aug 2025

## 0.32 mL

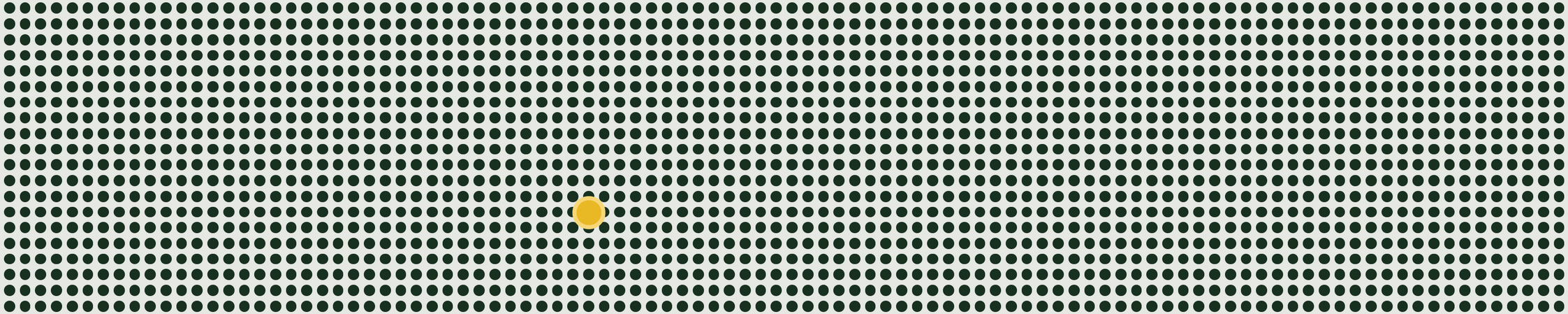
**A fifteenth of a teaspoon, per query**

The Gentle Singularity

> use in a little over one second, or a high-efficiency lightbulb would use in a couple of minutes. It also uses about 0.000085 gallons of water; roughly one fifteenth of a teaspoon.)

Sam Altman, blog, June 2025

# The largest answer is 2,000 times the smallest



Each dot is 0.26 mL, Google’s figure. All 2,000 together make the Washington Post’s 519 mL.

519 mL ÷ 0.26 mL ≈ 2,000

# The three figures measure different things

| | Washington Post | Google | Sam Altman |
|---|---|---|---|
| **Unit** | a 100-word email | the median text prompt | the average query |
| **Model, year** | GPT-4, 2024 | Gemini, 2025 | ChatGPT, 2025 |
| **Water counted** | cooling + power plants | on-site cooling only | not stated |
| **Statistic** | one figure | median | average |
| **Method shown** | yes, with UC Riverside | yes, arXiv paper | no |

The Post counts water used to generate the electricity; Google counts only cooling water at the data center; Altman does not say. The news coverage did not report which.

# Most claims reach the public without their source

# 0 of 634

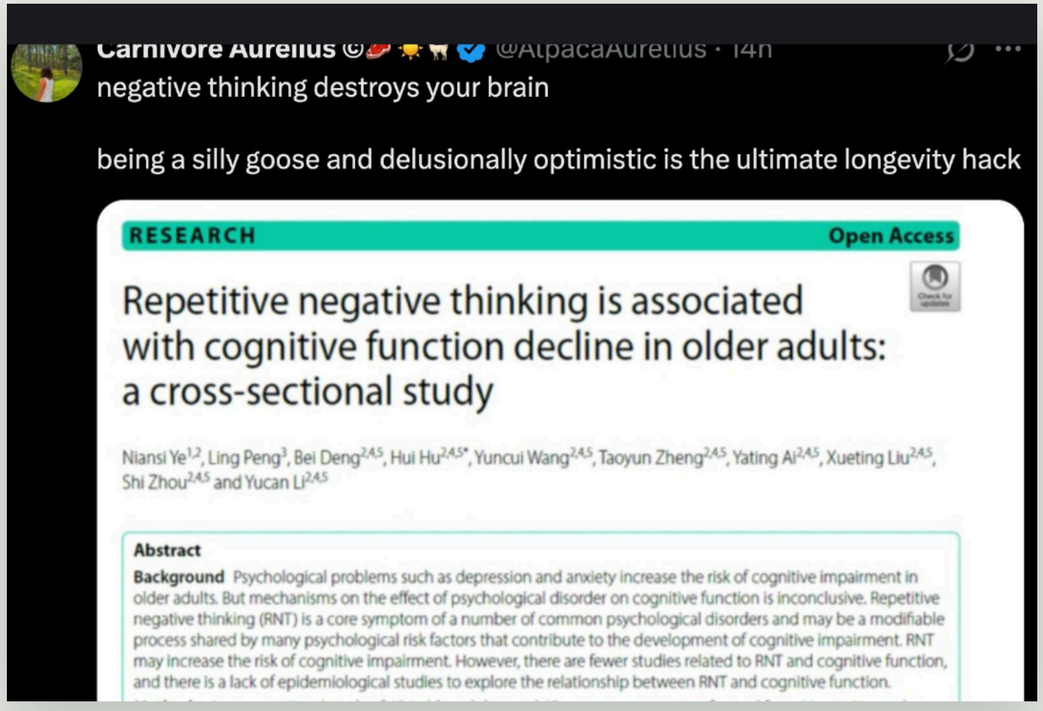
claims about AI water, in 77 news stories, link to the study they rest on.



No source, 44%

“A study found”, 53%

Findable, 2%

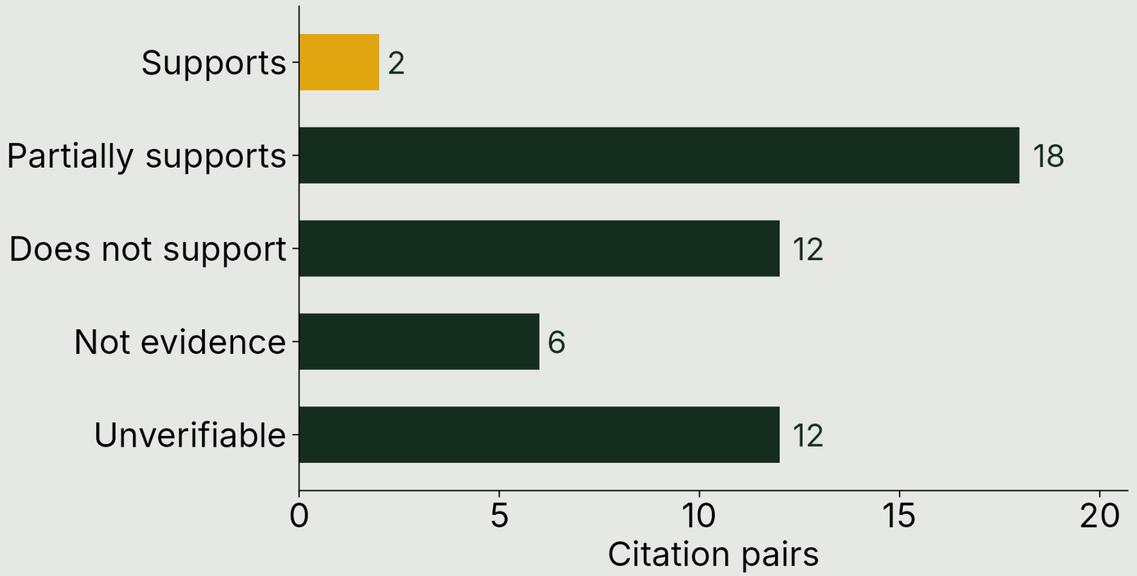



Moses showed me this post on day five. The paper studied 424 adults over 60 in one city, measured once; the most negative thinkers scored 1.6 points lower on a 30-point test.

# The drift starts inside the paper

We checked each of the 50 sentences in the Wuhan paper that cite another study against the study it cites.

Two are fully supported. Its line that negative thinking “depletes the brain’s limited resources” cites an eye-tracking experiment on attention.



# Pramāṇa, from Nyāya: the means by which knowledge is valid

## *pratyakṣa*

perception

The measurement: 424 people, one test, one number.

## *anumāna*

inference

What the design permits: association, not cause.

## *upamāna*

comparison

Every other study on the same question.

## *śabda*

testimony

The headline, the post, the press release.

Almost everything we know about science arrives as śabda. Pramana walks each piece of testimony back to the perception and inference underneath it, and sets it beside the comparisons.

# A claim is a set of fields, compared one by one

negative thinking destroys your brain

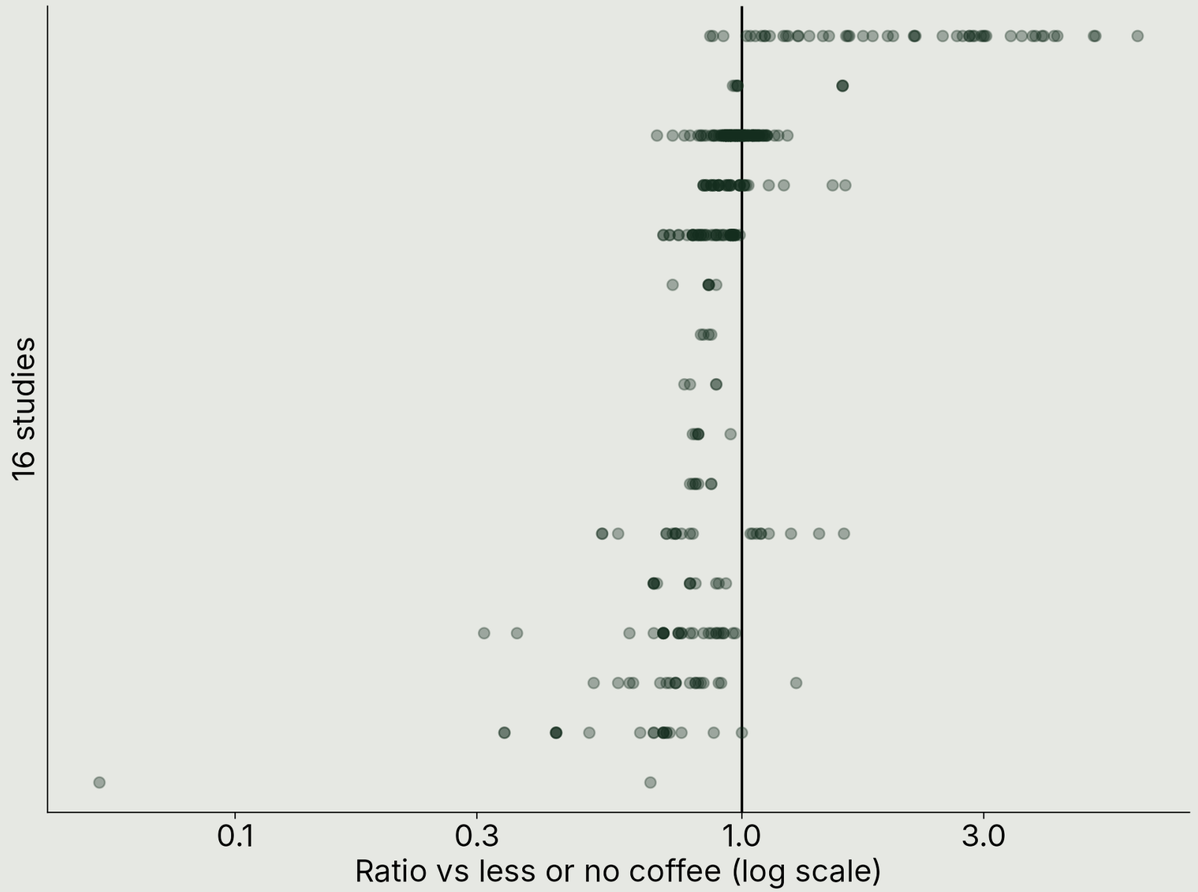
| Field | Paper | Post | Difference, fidelity lens v1 |
|---|---|---|---|
| **Subject** | repetitive negative thinking | negative thinking | none |
| **Outcome** | MoCA score, a 30-point test | your brain | measure read as organ |
| **Relation** | associated with | destroys | association read as cause |
| **Size** | 1.6 points lower | destroys | inflated |
| **Population** | 424 adults over 60, Wuhan | you | overgeneralized |
| **Certainty** | “further studies needed” | stated as fact | hedge dropped |

Subject and outcome are pinned to Wikidata IDs so the same idea matches across papers

# Coffee and mortality: 16 studies, 437 estimates

Each row is a study; each dot is one estimate it reports, for one dose, subgroup or model. Most sit just below 1.

A headline picks one dot. Pramana shows the whole field, with each dot linked to its sentence.



# Pramana app: search the evidence behind a claim

Paste a headline, a post or a sentence. The app breaks it into subject, outcome, population, effect size and certainty, and compares each with the papers that measured it.

It does not output true or false. It shows the numbers and where they came from.

**PRAMANA RECEIPT**

4 published answers to one question

QUESTION

How much water does one AI answer use?

| | **Washington Post**<br>with UC Riverside · 18 Sep 2024 | **Google**<br>Elsworth et al. · arXiv 2508.15734 · Aug 2025 | **Sam Altman**<br>The Gentle Singularity · Jun 2025 | **Ren et al.**<br>Making AI Less Thirsty · CACM |
|---|---|---|---|---|
| **ANSWER** | **519 mL**<br>PER 100-WORD EMAIL | **0.26 mL**<br>PER MEDIAN PROMPT | **0.32 mL**<br>PER AVERAGE QUERY | **10 to 50 mL**<br>PER RESPONSE |
| **Unit** | one 100-word email | median Gemini Apps text prompt | average ChatGPT query | one medium-length response |
| **Model and year** | GPT-4, 2024 | Gemini, 2025 | ChatGPT, 2025 | GPT-3, 2023 |
| **Water counted** | UC Riverside estimate | on-site cooling only | not stated | on-site cooling plus off-site power generation |
| **Statistic** | single figure | median | average | range, by place and time |
| **Method published** | analysis with UC Riverside | yes, arXiv paper | no | yes, peer reviewed |

**SPREAD, HIGHEST TO LOWEST** **519 ÷ 0.26 ≈ 2,000×**

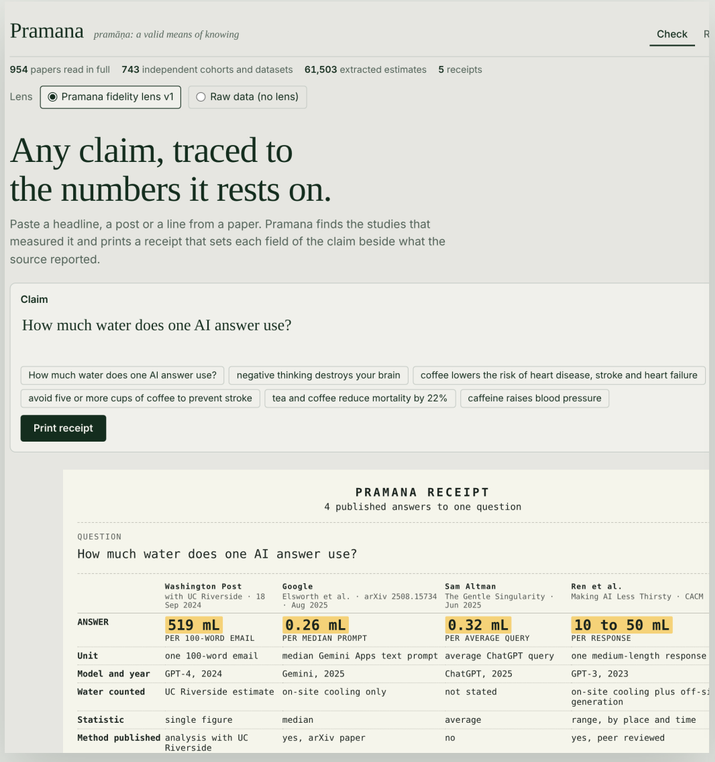
The four figures differ in unit, model, year and which water is counted.

**Underlying paper.** Ren et al.: GPT-3 drinks a 500 mL bottle for roughly 10 to 50 medium-length responses, depending on when and where it is deployed.

▸ Interpreted with Pramana fidelity lens v1 · adb · rules

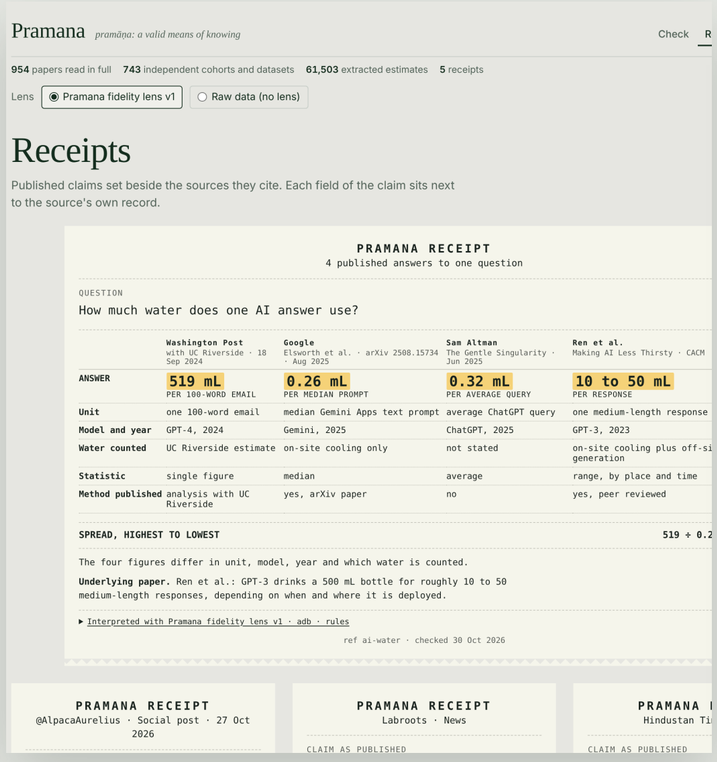
ref ai-water · checked 30 Oct 2026

# The app has three views



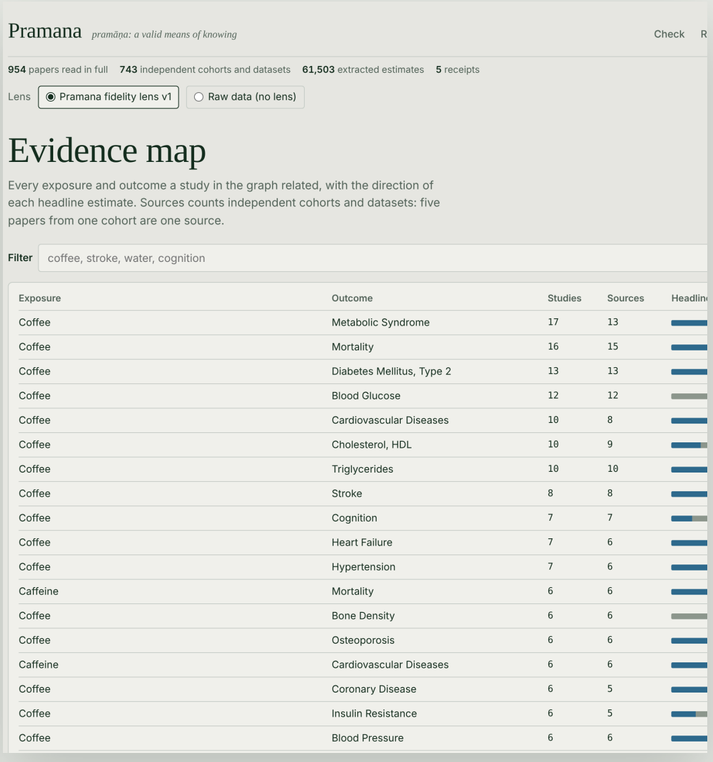

## Check

Paste any claim. See its receipt and the studies that measured it.




## Receipts

Published claims set against the sources they cite.




## Evidence map

Every exposure and outcome in the knowledge base, with the direction of each study's headline estimate.

# Pramana protocol: shared data, separate lenses

Contributors

### Run the pinned pipeline

Same prompts, model versions and schema, each pinned by hash. Contributors bring their own API keys.

Data layer, on Logos Storage

### What each paper reported

Estimates, intervals, populations and designs, each with a pointer to its sentence. Two independent extractions must agree. No verdicts.

Lenses

### Signed rules that interpret the data

A lens decides what counts as faithful, inflated or causal. Pramana's fidelity lens v1 is one. Anyone can publish another.

Readers

### Any client

Query a local copy through any lens, or none. Where two lenses disagree, the reader sees both.

Judgments never enter the data layer. People can argue about lenses because the facts underneath are shared.

# Architecture

| 1 · Sources | 2 · Read | 3 · Check | 4 · Know | 5 · Share |
|---|---|---|---|---|
| **PubMed Central**<br>full text, JATS XML | **Parse**<br>sections, tables, figures | **Validate**<br>schema, units, ranges | **Concepts**<br>MeSH and Wikidata IDs | **Snapshot**<br>SQLite, content-addressed |
| **OpenAlex**<br>open PDFs; abstracts otherwise | **Reader A**<br>Claude Haiku | **Agreement gate**<br>A vs B, field by field | **Evidence graph**<br>DuckDB, 1,055 papers | **Logos Storage**<br>retrieve by content ID |
| **News and posts**<br>RSS, pages, screenshots | **Reader B**<br>Laya, open model, our GPU | **Judge**<br>Claude Sonnet on disputes | **Lenses**<br>signed rules that interpret the data | **Basecamp module**<br>queries run locally |

**Pramana protocol**
Stages 1 to 4, pinned by hash. Anyone can run them with their own API keys. Lenses sit on top.

**Pramana app**
One reader of the shared knowledge base.

The pipeline records input and output hashes in an append-only ledger, and every extracted number keeps a pointer to the sentence, table or figure it came from.

Yellow: the Logos layer

# How a paper enters the knowledge base

1

Fetch the full text, tables and supplements

2

Two models extract every estimate independently

3

A judge settles where they disagree

4

Concepts pinned to Wikidata IDs

5

Claims matched to the evidence

**1,055**

papers, seven topics

**61,000+**

numbers, each linked to its sentence

**14¢**

per paper, all in

Readers: Claude Haiku and Laya, a small open model on our own GPU. Judge: Claude Sonnet.

λ Logos

# What Logos provides

Without it, every user would have to trust my server, my copy of the data and my logs.

| What a receipt needs | Logos does the work | Status |
|---|---|---|
| Proof it wasn’t edited after printing | Storage: every evidence snapshot has a content ID | built |
| Looking up a claim reveals nothing about you | Basecamp: the graph runs on your machine | built |
| Readers add the source a post left out | Messaging: notes travel peer to peer | next |
| Notes agreed across sides outrank one side | Blockchain: stake on bridging ratings | next |

# Demo

1

Paste the water question into Pramana in Basecamp

2

Read the receipt, field by field

3

Open the sentence each number came from

4

Match the snapshot’s content ID on Logos Storage

Then yours: bring any claim about coffee, aging, diabetes or AI water.

# Threat model

| Threat | Defense |
|---|---|
| A tampered snapshot | Content ID mismatch; the module refuses it |
| A model misreads a paper | Two readers, a judge, and every number points to its sentence |
| Someone logs what you check | Queries never leave your device |
| A biased verdict | Verdicts live in signed, swappable lenses, never in the data |
| Brigaded notes | Bridging ratings with stake (next) |

## Known limits

Open-access papers read in full; paywalled ones from the abstract only, and labeled so.

1,055 papers is a sample, not a library.

Extraction accuracy against human annotators: not yet measured.

English text only.

# Who can see what

| Who | What they can see |
|---|---|
| You | Your claim, your receipt, the full evidence graph on your own disk |
| Me | Nothing. No account, no server, no analytics |
| Logos Storage nodes | The public snapshot, never the questions asked of it |
| A note you choose to post | Signed with your key |

Lookups run on the user’s machine, so checking a health or political claim leaves no record anywhere. Two of the seven topics came from residents’ requests this week.

# Where a receipt changes a decision

### Newsrooms

A receipt beside every science claim, checked before it ships.

### Community notes

Notes that carry the numbers, not just a link.

### AI assistants

Answers that cite receipts instead of memory, through an MCP server.

### Regulators

Water permits for data centers in Rajasthan decided on counted water, not headlines.

### Courts

Every number in an expert report traced to its source.

### Journals and reviewers

A citation audit of each manuscript before it is accepted.

### Funders

A map of where evidence is thin, to decide which replications to pay for.

### Prediction markets

Questions that resolve on the evidence record.

# Users after the residency

### Science desks and fact-checkers

Paste a line before it ships. The receipt goes in the footnote.

### Researchers writing reviews

The graph is the first pass of a systematic review, at 14¢ a paper instead of a month of reading.

### Contributors

Labs and journal clubs add the papers in their field, with their own API keys.

Feedback: disputes filed by readers become labeled examples for retraining the open extraction model. An MCP server lets AI agents query the same knowledge base.

# Next three milestones

1 · Release

Protocol spec and contributor tool published. Basecamp app on testnet. Snapshots on Logos Storage. 10,000 papers.

2 · Pilot

Two desks use it for a month. Accuracy measured against human annotators. A public report of what broke.

3 · Adoption

Agreement checking across independent contributors. Notes over Messaging. MCP server. Self-audit. A named operator running a node.

# A public record of what the evidence says, grown by anyone and owned by no one.

*Next time someone tells you how much water your question used, ask which water they counted.*

Amruth Deepak Bhat · github.com/adb3502/pramana

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